

```
In [1]: import pandas as pd
```

```
In [2]: import numpy as np
```

```
In [3]: df = pd.read_csv("student_admission_record_dirty.csv")
```

```
In [5]: df.head(10)
```

```
Out[5]:
```

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
0	Shehroz	24.0	Female	50.0	68.90	Quetta	Rejected
1	Waqar	21.0	Female	99.0	60.73	Karachi	NaN
2	Bushra	17.0	Male	89.0	NaN	Islamabad	Accepted
3	Aliya	17.0	Male	55.0	85.29	Karachi	Rejected
4	Bilal	20.0	Male	65.0	61.13	Lahore	NaN
5	Murtaza	23.0	Female	NaN	NaN	Islamabad	Accepted
6	Asad	18.0	Male	NaN	97.31	Multan	Accepted
7	Rabia	20.0	Female	82.0	55.67	Lahore	Accepted
8	Rohail	17.0	Male	64.0	NaN	Karachi	Accepted
9	Kamran	18.0	Male	53.0	98.98	Multan	Rejected

```
In [6]: df.tail()
```

```
Out[6]:
```

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
152	Ali	19.0	Female	85.0	78.09	Quetta	Accepted
153	Bilal	17.0	Female	81.0	84.40	Islamabad	Rejected
154	Fatima	21.0	Female	98.0	50.86	Multan	Accepted
155	Shoaib	-1.0	Male	91.0	80.12	Quetta	Accepted
156	Maaz	17.0	Male	88.0	86.85	Lahore	Accepted

```
In [8]: df.shape
```

```
Out[8]: (157, 7)
```

```
In [9]: df.columns
```

```
Out[9]: Index(['Name', 'Age', 'Gender', 'Admission Test Score',
              'High School Percentage', 'City', 'Admission Status'],
              dtype='object')
```

```
In [13]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 157 entries, 0 to 156
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Name                  147 non-null   object
1   Age                   147 non-null   float64
2   Gender                147 non-null   object
3   Admission Test Score  146 non-null   float64
4   High School Percentage 146 non-null   float64
5   City                  147 non-null   object
6   Admission Status      147 non-null   object
dtypes: float64(3), object(4)
memory usage: 8.7+ KB
```

```
In [15]: df.isnull()
```

Out[15]:

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	True
2	False	False	False	False	True	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	True
...	...	...	...	...	...	...	...
152	False	False	False	False	False	False	False
153	False	False	False	False	False	False	False
154	False	False	False	False	False	False	False
155	False	False	False	False	False	False	False
156	False	False	False	False	False	False	False

157 rows × 7 columns

```
In [16]: df.isnull().sum()
```

```
Out[16]: Name          10
        Age           10
        Gender        10
        Admission Test Score  11
        High School Percentage  11
        City          10
        Admission Status  10
        dtype: int64
```

```
In [17]: df.describe()
```

```
Out[17]:
```

	Age	Admission Test Score	High School Percentage
count	147.000000	146.000000	146.000000
mean	19.680272	77.657534	75.684726
std	4.540512	16.855343	17.368014
min	-1.000000	-5.000000	-10.000000
25%	18.000000	68.250000	65.052500
50%	20.000000	79.000000	77.545000
75%	22.000000	89.000000	88.312500
max	24.000000	150.000000	110.500000

```
In [18]: df.ndim
```

```
Out[18]: 2
```

```
In [21]: df.dtypes
```

```
Out[21]: Name          object
        Age          float64
        Gender        object
        Admission Test Score  float64
        High School Percentage  float64
        City          object
        Admission Status  object
        dtype: object
```

```
In [23]: df['Age'] = df['Age'].astype('Int64')
```

```
In [32]: df.dtypes
```

```
Out[32]: Name          object
        Age          Int64
        Gender        object
        Admission Test Score  float64
        High School Percentage  float64
        City          object
        Admission Status  object
        dtype: object
```

```
In [37]: df.isnull().sum()
```

```
Out[37]: Name                10
Age                  0
Gender              10
Admission Test Score 11
High School Percentage 0
City                10
Admission Status     10
dtype: int64
```

```
In [34]: df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
In [36]: df['High School Percentage']=df['High School Percentage'].fillna(df['High School Pe
```

```
In [39]: df.head()
```

```
Out[39]:
```

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
0	Shehroz	24	Female	50.0	68.900000	Quetta	Rejected
1	Waqar	21	Female	99.0	60.730000	Karachi	NaN
2	Bushra	17	Male	89.0	75.684726	Islamabad	Accepted
3	Aliya	17	Male	55.0	85.290000	Karachi	Rejected
4	Bilal	20	Male	65.0	61.130000	Lahore	NaN

```
In [42]: df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})
```

```
In [45]: df.head()
```

```
Out[45]:
```

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
0	Shehroz	24	1.0	50.0	68.900000	Quetta	0.0
1	Waqar	21	1.0	99.0	60.730000	Karachi	NaN
2	Bushra	17	0.0	89.0	75.684726	Islamabad	1.0
3	Aliya	17	0.0	55.0	85.290000	Karachi	0.0
4	Bilal	20	0.0	65.0	61.130000	Lahore	NaN

```
In [44]: df['Admission Status'] = df['Admission Status'].map({'Rejected': 0, 'Accepted': 1})
```

```
In [46]: df.head()
```

Out[46]:

	Name	Age	Gender	Admission Test Score	High School Percentage	City	Admission Status
0	Shehroz	24	1.0	50.0	68.900000	Quetta	0.0
1	Waqar	21	1.0	99.0	60.730000	Karachi	NaN
2	Bushra	17	0.0	89.0	75.684726	Islamabad	1.0
3	Aliya	17	0.0	55.0	85.290000	Karachi	0.0
4	Bilal	20	0.0	65.0	61.130000	Lahore	NaN

In []: